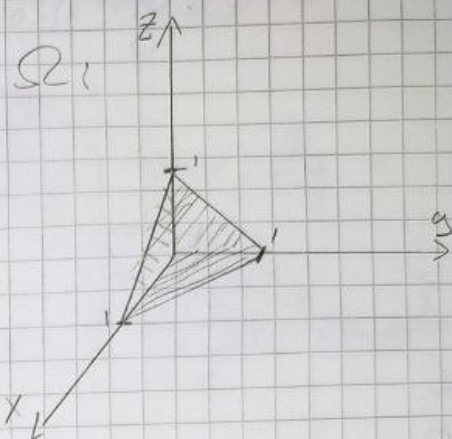


3522

⑬

$$\iiint_{\Omega} \frac{dx dy dz}{(x+y+z+1)^3}$$

$$\Omega: \begin{cases} x=0 \\ y=0 \\ z=0 \\ x+y+z=1 \end{cases}$$



$$x \in [0; 1]$$

$$0 \leq y \leq 1-x$$

$$0 \leq z \leq 1-x-y$$

$$\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} \frac{dz}{(x+y+1+z)^3} = \int_0^1 dx \int_0^{1-x} dy \cdot \left( \frac{(x+y+1+z)^{-3+1}}{-3+1} \right) \Big|_0^{1-x-y} =$$

$$= \int_0^1 dx \int_0^{1-x} -\frac{1}{2} dy \cdot \left( \frac{1}{(x+y+1+1-x-y)^2} - \frac{1}{(x+y+1)^2} \right) = \int_0^1 dx \int_0^{1-x} \left( \frac{1}{2(x+y+1)^2} - \frac{1}{8} \right) dy = \int_0^1 \frac{1}{2} dx \cdot \left( \frac{(x+y+1)^{-2+1}}{-2+1} - \frac{1}{4} y \right) \Big|_0^{1-x} = \frac{1}{2} \int_0^1 dx \left( -\frac{1}{(x+1+1-x)^1} + \frac{1}{4}(x+1) - \frac{1}{4}(1-x-0) \right) = \frac{1}{2} \int_0^1 dx \cdot \left( \frac{1}{x+1} - \frac{1}{2} + \frac{1}{4}x - \frac{1}{8} \right) = \frac{1}{2} \int_0^1 \left( \frac{1}{x+1} + \frac{1}{4}(x-1-2) \right) dx =$$

$$= \frac{1}{2} \int_0^1 \frac{1}{x+1} + \frac{1}{4}(x-3) dx = \frac{1}{2} \left( \ln(x+1) + \frac{1}{4} \cdot \frac{x^2}{2} - \frac{3}{4}x \right) \Big|_0^1 =$$

$$= \frac{1}{2} \left( \ln 2 + \frac{1}{8} - \frac{3}{4} - \ln(0+1) + \frac{1}{8} \cdot 0 - \frac{3}{4} \cdot 0 \right) = \frac{1}{2} \ln 2 - \frac{5}{16}$$

3523

$$\iiint_{\Omega} xy \, dx dy dz$$

$$\Omega: \begin{cases} z=xy - \text{гиперболическая параболоид} \\ x+y=1 \\ z=0 \quad (z \geq 0) \end{cases}$$

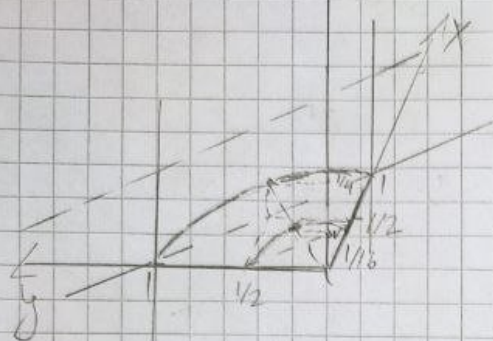


②  $\Omega$ :

$z$

$$\begin{cases} z = xy \\ x + y = 1 \end{cases} \quad \begin{cases} y = 1 - x \\ z = x(1 - x) \end{cases}$$

$$\begin{matrix} x & 1 & 0 & 1/2 \\ y & 0 & 1 & 1/2 \\ z & 0 & 0 & 1/4 \end{matrix}$$



Для построения  $x + y = \frac{1}{2}$ :

$$\begin{cases} z = xy \\ x + y = \frac{1}{2} \end{cases} \Rightarrow \begin{cases} xy = \frac{1}{2} - x \\ z = x(\frac{1}{2} - x) \end{cases} \quad \begin{matrix} x & 1/2 & 0 & 1/4 \\ y & 0 & 1/2 & 1/4 \\ z & 0 & 0 & 1/16 \end{matrix}$$

$\Omega$ :

$$0 \leq x \leq 1$$

$$0 \leq y \leq 1 - x$$

$$0 \leq z \leq xy$$

$$\iiint_{\Omega} xy \, dx \, dy \, dz = \int_0^1 dx \int_0^{1-x} dy \int_0^{xy} xy \, dz =$$

$$= \int_0^1 dx \int_0^{1-x} (xy \cdot z) \Big|_0^{xy} dy = \int_0^1 dx \int_0^{1-x} x^2 y^2 dy = \int_0^1 dx \cdot x^2 \cdot \frac{y^3}{3} \Big|_0^{1-x} =$$

$$= \int_0^1 x^2 dx \cdot \frac{1}{3} (1-x)^3 = \frac{1}{3} \int_0^1 (x^2 - 3x \cdot x^2 + 3x^2 \cdot x^2 - x^3 \cdot x^2) dx =$$

$$= \frac{1}{3} \left( \frac{x^3}{3} - 3 \frac{x^4}{4} + 3 \frac{x^5}{5} - \frac{x^6}{6} \right) \Big|_0^1 = \frac{1}{3} \left( \frac{1}{3} - \frac{3}{4} + \frac{3}{5} - \frac{1}{6} \right) =$$

$$= \frac{1}{3} \cdot \frac{1}{3} \left( 1 - \frac{9}{4} + \frac{9}{5} - \frac{1}{2} \right) = \frac{1}{9} \cdot \frac{1}{20} = \frac{1}{180}$$

3524

$$\iiint_{\Omega} y \cos(z+x) \, dx \, dy \, dz$$

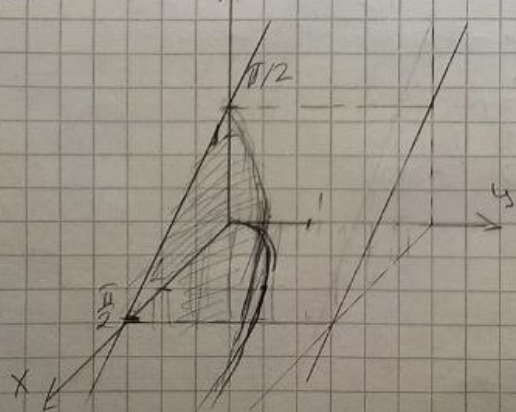
$z$

$$\Omega: y = \sqrt{x} \text{ и } \pi/2 \Rightarrow x \geq 0$$

$$y = 0$$

$$z = 0$$

$$x + z = \frac{\pi}{2}$$



$$\Omega: 0 \leq x \leq \frac{\pi}{2}$$

$$0 \leq y \leq \sqrt{x}$$

$$0 \leq z \leq \frac{\pi}{2} - x$$



$$\iiint_{\Omega} y \cos(z+x) dx dy dz = \int_0^{\pi/2} dx \int_0^{\sqrt{x}} dy \int_0^{\pi/2-x} \cos(z+x) y dz = \quad (3)$$

$$= \int_0^{\pi/2} dx \int_0^{\sqrt{x}} y dy \cdot \sin(z+x) \Big|_0^{\pi/2-x} = \int_0^{\pi/2} dx \int_0^{\sqrt{x}} y (1 - \sin x) dy =$$

$$= \int_0^{\pi/2} (1 - \sin x) dx \cdot \frac{y^2}{2} \Big|_0^{\sqrt{x}} = \int_0^{\pi/2} (1 - \sin x) dx \cdot \frac{x}{2} = \frac{1}{2} \int_0^{\pi/2} (x - x \sin x) dx$$

Вычисляем  $\int (x - x \sin x) dx$ .

$$\int (x - x \sin x) dx = \int x dx - \int x \sin x dx = \frac{x^2}{2} + \int x d \cos x =$$

$$\frac{x^2}{2} + x \cos x - \int \cos x dx = \frac{x^2}{2} + x \cos x - \sin x$$

$$\frac{1}{2} \int_0^{\pi/2} (x - x \sin x) dx = \frac{1}{2} \left( \frac{x^2}{2} + x \cos x - \sin x \right) \Big|_0^{\pi/2} = \frac{1}{2} \left( \frac{\pi^2}{2 \cdot 4} + 2^0 \right)$$

$$\cdot \cos \frac{\pi}{2} - \sin \frac{\pi}{2} - \frac{1}{2} \cdot 0 - 0 \cos 0 + \sin 0) = \underline{\underline{\pi^2/16 - 1/2}}$$

3537

Вычисляем с переходом в полярные координаты

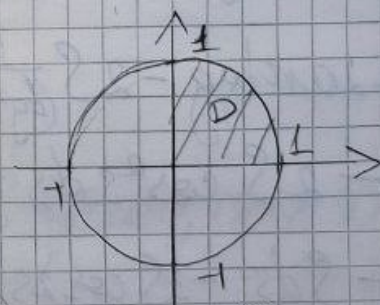
$$\iint_D \sqrt{\frac{1-x^2-y^2}{1+x^2+y^2}} dx dy, \quad D: \begin{cases} x^2+y^2 \leq 1 \\ x \geq 0 \\ y \geq 0 \end{cases}$$

$$x = r \cos \varphi$$

$$y = r \sin \varphi$$

$$\iint_D \sqrt{\frac{1-x^2-y^2}{1+x^2+y^2}} dx dy = \int_0^{\pi/2} d\varphi \int_0^1 \sqrt{\frac{1-r^2}{1+r^2}} r dr =$$

$$= \int_0^{\pi/2} \frac{1}{2} d\varphi \int_0^1 \sqrt{\frac{1-r^2}{1+r^2}} dr^2 = \left[ \begin{matrix} E = r^2 \\ E_1 = \pi^2/4 \\ E_0 = 0 \end{matrix} \right] =$$



$$0 \leq \varphi \leq \frac{\pi}{2}$$

$$r^2 \leq 1, r \geq 0$$

$$0 \leq r \leq 1$$



$$④ = \int_0^{\pi/2} \frac{1}{2} d\varphi \int_0^1 \sqrt{\frac{1-z}{1+z}} dz = \left[ z = \frac{1-t}{1+t} \Rightarrow z = \frac{1-t}{1+t} \right. \\ \left. dz = \frac{-t-1-(1+t)}{(1+t)^2} dt = \frac{-2dt}{(1+t)^2} \right] = \\ z_1 = 1 \Rightarrow t_1 = \frac{1-1}{1+1} = 0 \\ z_0 = 0 \Rightarrow t_0 = \frac{1-0}{1+0} = 1$$

$$= \int_0^{\pi/2} \frac{1}{2} d\varphi \int_1^0 \sqrt{t} \cdot \frac{-2dt}{(1+t)^2} = \int_0^{\pi/2} d\varphi \int_0^1 \frac{\sqrt{t} dt}{(1+t)^2} = \left[ g = \sqrt{t} \quad g_1 = t_1 \right. \\ \left. dt = 2g dg \quad g_0 = t_0 \right]$$

$$= \int_0^{\pi/2} d\varphi \int_0^1 \frac{g \cdot 2g dg}{(g^2+1)^2} \quad \text{Bernoulli} \left( \int \frac{2g^2 dg}{(g^2+1)^2} \right)$$

$$2 \int \frac{g^2 dg}{(g^2+1)^2} = 2 \int \left( \frac{A}{g^2+1} + \frac{B}{(g^2+1)^2} \right) dg = 2 \int \left( \frac{1}{g^2+1} - \frac{1}{(g^2+1)^2} \right) dg \quad ②$$

$$\frac{g^2}{(g^2+1)^2} = \frac{A}{g^2+1} + \frac{B}{(g^2+1)^2}$$

$$g^2 = A(g^2+1) + B$$

$$Ag^2 + A + B = g^2 \Rightarrow \begin{cases} A=1 \\ A+B=0 \end{cases} \Rightarrow \begin{cases} A=1 \\ B=-1 \end{cases}$$

$$\textcircled{2} 2 \int \frac{dg}{g^2+1} - 2 \int \frac{dg}{(g^2+1)^2} = 2 \cdot \arctan g - 2 \cdot \int \frac{dg}{(g^2+1)^2} = \left[ g = \tan s \right. \\ \left. s = \arctan g \right. \\ \left. dg = \frac{ds}{\cos^2 s} \right] = \\ = 2 \arctan g - 2 \int \frac{1}{(\tan^2 s + 1)^2} \cdot \frac{ds}{\cos^2 s} = 2 \arctan g -$$

$$- 2 \int \cos^2 s ds = 2 \arctan g - 2 \int \frac{1+\cos 2s}{2} ds = 2 \arctan g -$$

$$- \int ds - \int \cos 2s ds = 2 \arctan g - s - \frac{1}{2} \sin 2s.$$

$$\int_0^{\pi/2} d\varphi \int_0^1 \frac{2g^2 dg}{(g^2+1)^2} = \int_0^{\pi/2} d\varphi \left( 2 \arctan g \Big|_0^1 - \left( s + \frac{1}{2} \sin 2s \right) \Big|_{\arctan 0}^{\arctan 1} \right) =$$

$$= \int_0^{\pi/2} d\varphi \cdot \left( 2 \left( \frac{\pi}{4} - 0 \right) - \left( \frac{\pi}{4} + \frac{1}{2} \cdot 1 - 0 - \frac{1}{2} \cdot 0 \right) \right) = \int_0^{\pi/2} \left( \frac{\pi}{4} - \frac{1}{2} \right) d\varphi =$$



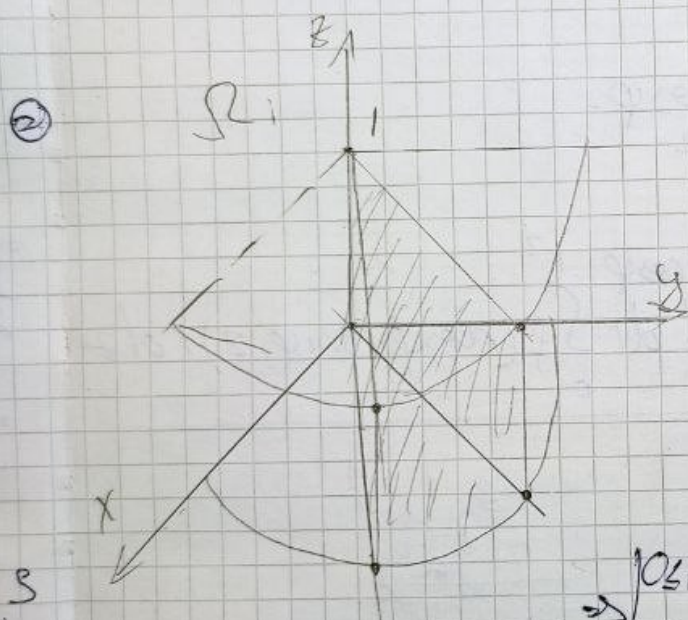
$$= \frac{\pi-2}{4} \cdot \varphi \Big|_0^{\pi/2} = \frac{\pi-2}{4} \cdot \frac{\pi}{2} = \frac{\pi(\pi-2)}{8}$$

⑤

3547

Перейти в  $\iiint_{\Omega} f(x,y,z) dx dy dz$  к цилиндрическим координатам  $r, \varphi, z$  или сферическим координатам  $r, \theta, \varphi$  и расставить пределы интегрирования.

$\Omega$  - область в I октанте :  $x^2 + y^2 = R^2$  цилиндр



$$\begin{cases} z=0 \\ y=x \\ z=1 \\ y=x\sqrt{3} \end{cases}$$

$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \\ z = z \end{cases} \Rightarrow \begin{cases} r^2 = R^2 \\ z=0 \\ z=1 \\ r \cos \varphi = r \sin \varphi \\ r \sin \varphi = r \cos \varphi \sqrt{3} \end{cases}$$

$$\Rightarrow \begin{cases} 0 \leq r \leq R \\ 0 \leq z \leq 1 \\ \frac{\pi}{4} \leq \varphi \leq \frac{\pi}{3} \end{cases}$$

$$\iiint_{\Omega} f(x,y,z) dx dy dz = \int_{\pi/4}^{\pi/3} d\varphi \int_0^R dr \int_0^1 f(r \cos \varphi, r \sin \varphi, z) r dz$$

3548

$\Omega$  :  $\begin{cases} x^2 + y^2 = 2x \text{ цилиндр} \\ z=0 \\ z = x^2 + y^2 \text{ параболоид} \end{cases}$

rectg 1  
rectg 0

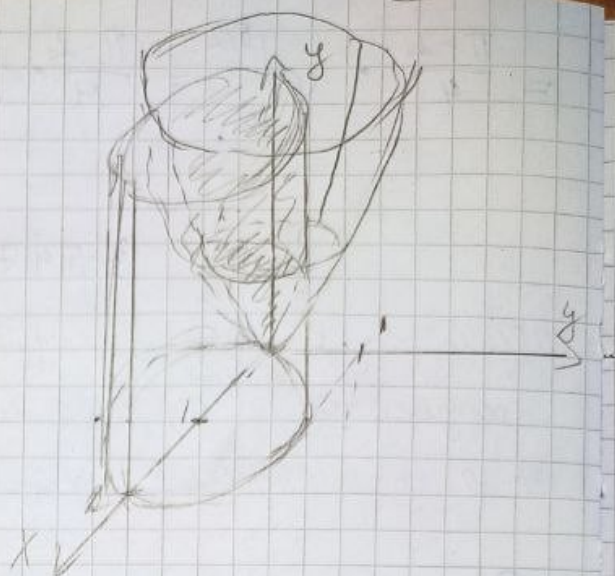


$$⑥ \quad x^2 + y^2 = 2x \Leftrightarrow (x-1)^2 + y^2 = 1$$

$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \\ z = z \end{cases}$$

$$\Omega: \begin{cases} r^2 \cos^2 \varphi + r^2 \sin^2 \varphi = 2r \cos \varphi \\ z = 0 \\ r^2 \cos^2 \varphi + r^2 \sin^2 \varphi = z \end{cases}$$

$$\begin{cases} r = 2 \cos \varphi & -\frac{\pi}{2} \leq \varphi \leq \frac{\pi}{2} \\ r^2 = z & 0 \leq r \leq 2 \cos \varphi \\ z = 0 & 0 \leq z \leq r^2 \end{cases}$$



$$\iiint_{\Omega} f(x, y, z) dx dy dz = \int_{-\pi/2}^{\pi/2} d\varphi \int_0^{2 \cos \varphi} dr \int_0^{r^2} f(r \cos \varphi, r \sin \varphi, z) r dz$$

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Вычислить с переходом

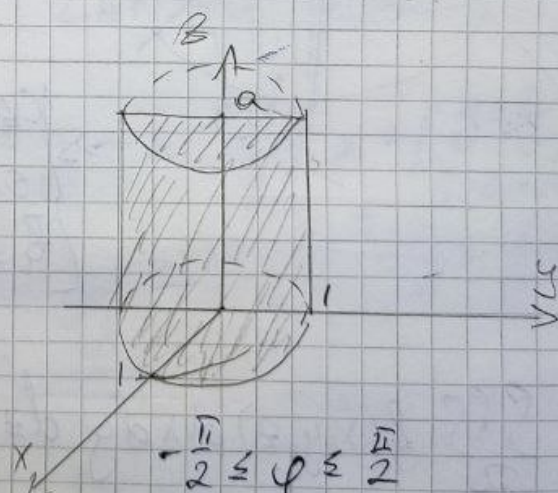
$$\int_0^1 dx \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy \int_0^a dz \quad (\Rightarrow)$$

$$D: \quad x \in [0; 1]$$

$$-\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2} \Rightarrow y^2 = 1-x^2$$

$$0 \leq z \leq a$$

$$x^2 + y^2 = 1$$



$$-\frac{\pi}{2} \leq \varphi \leq \frac{\pi}{2}$$

$$0 \leq r^2 \leq 1 \quad r \in [0; 1]$$

$$0 \leq z \leq a$$

$$\begin{aligned} (\Rightarrow) \int_{-\pi/2}^{\pi/2} d\varphi \int_0^1 dr \int_0^a r dz &= \int_{-\pi/2}^{\pi/2} d\varphi \int_0^1 r dr \cdot a = \int_{-\pi/2}^{\pi/2} a d\varphi \cdot \frac{1}{2} = \frac{a}{2} \left( \frac{\pi}{2} + \frac{\pi}{2} \right) = \frac{\pi a}{2} \end{aligned}$$